Family of parallel lines: Family of parallel lines to $ax+by+c=0$ is of the form $ax+by+\lambda=0$. For different values of λ (call it *lambda*), we get different lines parallel to $ax+by+c=0$.

Family of perpendicular lines: Family of perpendicular lines to $ax+by+c=0$ is of the form $bx-ay+\lambda=0$. For different values of λ, we get different lines perpendicular to $ax+by+c=0$.

6.4.7 Two parameters families

(iii) when both m and b are arbitrary

Suppose both m and b are arbitrary constants in $y = mx + b$, we cannot visualize the family easily in graph. But some cases like $y - y_1 = m(x - x_1)$ for different real values of m, a family of lines can be visualized. Suppose the slope m takes -2, -4, $\frac{1}{3}$, 1 and 3, lines which pass through the fixed point (x_1, y_1) except the vertical line $x = x_1$ as shown in the diagram.

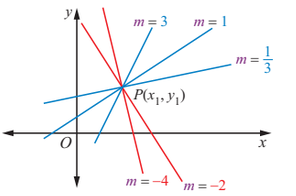


Figure 6.42

6.4.8 The family of equation of straight lines through the point of intersection of the two given lines.

Let $L_1 \equiv a_1x + b_1y + c_1 = 0$ and $L_2 \equiv a_2x + b_2y + c_2 = 0$,

be the equation of two given lines. The family of equations of straight lines through the point of intersection of the above lines is $L_1+\lambda L_2 = 0$ where λ is a parameter. That is, for different real values of λ we get different equations.

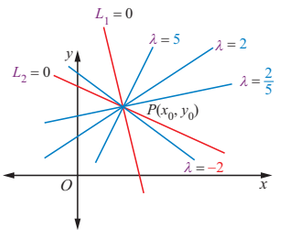


Figure 6.43

Example 6.29 Find the equation of the line through the intersection of the lines $3x+2y+5=0$ and $3x-4y+6=0$ and the point (1,1).

Solution:

The family of equations of straight lines through the point of intersection of the lines is of the form $(a_1x + b_1y + c_1) + \lambda(a_2x + b_2y + c_2) = 0$

That is, $(3x + 2y + 5) + \lambda(3x - 4y + 6) = 0$

Since the required equation passes through the point (1,1), the point satisfies the above equation

Therefore $\{3 + 2(1) + 5\} + \lambda\{3(1) - 4(1) + 6\} = 0 \Rightarrow \lambda = -2$

Substituting $\lambda = -2$ in the above equation we get the required equation as $3x - 10y + 7 = 0$ (verify the above problem by using two points form)

Example 6.30 Suppose the Government has decided to erect a new Electrical Power Transmission Substation to provide better power supply to two villages namely A and B. The substation has to to be on the line l. The distances of villages A and B from the foot of the perpendiculars P and Q on the line l are 3 km and 5 km respectively and the distance between P and Q is 6 km. (i) What is the smallest length of cable required to connect the two villages. (ii) Find the equations of the cable lines that connect the power station to two villages. (Using the knowledge in conjunction with the principle of reflection allows for approach to solve this problem.)

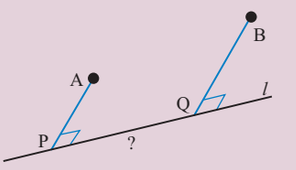


Figure 6.44